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ARTICLE

Plasma 11-Ketotestosterone in Individual Age-1 Spring Chinook Salmon Males Accurately Predicts Age-2 Maturation Status

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Abstract

High rates of precocious male maturation of spring Chinook Salmon *Oncorhynchus tshawytscha* as 2-year-old minijacks have been observed in Columbia River basin hatchery programs. Previously, minijack rates have been assessed by lethal sampling during the spring prior to release of the age-1 smolts and measurement of plasma levels of 11-ketotestosterone (11-KT), the major androgen in salmonids. Analysis of plasma 11-KT levels has been used for separation of male fish into two distinct groups: low levels for immature males and high levels for precociously maturing minijacks. However, no published studies have confirmed how accurately plasma 11-KT levels measured in the spring prior to smolt release correspond to the maturation status of individual males the following fall spawning season. We utilized both lethal and nonlethal sampling to determine when to sample hatchery-reared juveniles to obtain clear separation between immature and precociously maturing males and PIT-tagged fish to determine whether plasma 11-KT levels in the spring accurately predicted an individual's maturation status in the fall. The distribution of plasma 11-KT values from fish sampled in February was not significantly bimodal, whereas that from fish sampled in April was strongly bimodal. Plasma 11-KT levels measured nonlethally in April accurately predicted the maturation status of 99% of male fish, confirmed via the gonadosomatic index obtained the following September. Plasma 11-KT levels increased from April to September in both immature and maturing males, although the percent increase in maturing males was 10 times greater than that experienced by the non-maturing smolts. Our findings support the use of spring plasma 11-KT level as an indicator of maturation status in Chinook Salmon juveniles, provided that an appropriate sampling time is selected.

In anadromous salmonids, precocious maturation during freshwater rearing has been observed in a proportion of male fish (Thorpe 1994; Taranger et al. 2010; Sloat et al. 2014). Precocious male maturation occurs in spring Chinook Salmon *Oncorhynchus tshawytscha*, both as age-1 precocial parr (age being calculated from the time of egg fertilization) and as age-2 "minijacks" (Gebhards 1960; Foote et al. 1991; Mullan et al. 1992; Clarke and Blackburn 1994; Zimmerman et al. 2003). Although the frequency of males maturing precociously is typically low in wild populations, this proportion can be elevated among fish reared in hatcheries (Gebhards 1960; Mullan et al. 1992; Larsen et al. 2004; Harstad et al. 2014). In wild populations, precocious males may successfully fertilize eggs from large anadromous females by adopting a "sneaker" strategy, which maintains the trait in the population; however, frequency-dependent selection keeps the proportion of precocious males low (Gross 1985; Hankin et al. 2009; Berejikian et al. 2010; Schroder et al. 2012). The increased occurrence of precociously maturing males in hatcheries is associated with the accelerated growth and higher energy reserves in juvenile fish in the hatchery environment relative to juveniles in the natural environment (Rowe and Thorpe 1990; Thorpe et al. 1990; Rowe et al. 1991). However, while observations of survival after maturation in precocious parr have been documented in spring Chinook Salmon (Rich 1920; Gebhards 1960; Bernier et al. 1993), it is presumed that minijacks die after the spawning season in the natural environment (Gebhards 1960). Therefore, production of large numbers of non-sea-going, precociously maturing males among hatchery-reared age-1+ smolts (approximately 18 months postfertilization) undermines the objective of hatchery programs to produce increased numbers of ocean-returning adults for harvest augmentation, supplementation, and conservation of wild salmonid populations (Beckman and Larsen 2005; Spangenberg et al. 2015).

The incidence of minijacks in wild stocks of spring Chinook Salmon is poorly characterized but is thought to be less than 5% (Larsen et al. 2004). In contrast, the proportion of juvenile males maturing as minijacks in hatchery populations of Chinook Salmon ranges from 11% to greater than 80% (Foote et al. 1991; Mullan et al. 1992; Clarke and Blackburn 1994; Zimmerman et al. 2003; Harstad et al. 2014). Conservation hatchery programs for spring Chinook Salmon, which incorporate natural-origin broodstock and aim to produce fish with genetic characteristics similar to those of the native stock, have the potential benefits of reducing the short-term risk of extinction, speeding recovery, reseeding vacant habitat, and increasing harvest opportunity (Fast et al. 2015). However, these integrated programs also produce minijacks at rates higher than segregated hatchery programs that only use hatchery-origin adults for broodstock (Harstad et al. 2014; Spangenberg et al. 2014, 2015). Thus, it appears that both genetic background and the rearing environment influence precocious maturation (Thorpe and Morgan 1980; Saunders 1986; Thorpe et al. 1990; Rowe et al. 1991; Larsen et al. 2006; Mangel and Satterthwaite 2008).

The maturation process is regulated by the reproductive endocrine axis in salmonids, which consists of the brain, pituitary gland, and gonads (Okuzawa 2002; Taranger et al. 2010). Initiation of maturation for fall-spawning species is thought to occur during a critical period a full year prior to the spawning season (Mangel 1994; Thorpe 1994; Berglund 1995; Silverstein et al. 1998). In male Chinook Salmon, concurrent increases in the proportion of transitional spermatogonia in the testes and plasma levels of 11-ketotestosterone (11-KT), the principal testicular androgen

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in teleost fishes, have been identified over a time period that is consistent with the proposed critical period for initiation of maturation (Campbell et al. 2003). However, a clearly bimodal frequency distribution of 11-KT levels that can reliably distinguish maturing from non-maturing males is not observed until the following spring (Campbell et al. 2003; Larsen et al. 2006), which is also the time of a second proposed critical period during which maturation can be either arrested or allowed to continue (Mangel 1994; Thorpe 1994; Berglund 1995; Silverstein et al. 1998). Fish that continue maturation after the second critical period are thought to be committed to completing the maturation process by the subsequent fall spawning season (Thorpe et al. 1998). Both initiation and continuation of maturation depend upon the energy stores of the individual during these critical periods (Thorpe 1994). Previous research has shown that a fish will mature if the rate of acquisition or attainment of a particular energy-related trait (e.g., growth rate, adiposity, or size) during a critical period exceeds a genetically determined threshold. The physiological mechanisms involved in these decisions are not yet clear (Mangel and Satterthwaite 2008; Taranger et al. 2010; Good and Davidson 2016). However, it has been demonstrated that the minijack rate increases with increased feeding and energy reserves in hatchery-reared Chinook Salmon juveniles (Larsen et al. 2006; Spangenberg et al. 2014).

To date, studies assessing the minijack rate in hatchery-reared spring Chinook Salmon have involved lethally sampling a subset of fish prior to smolt release, measuring plasma 11-KT levels, and reporting the minijack rate as the proportion of male fish with elevated 11-KT (Larsen et al. 2004, 2013; Harstad et al. 2014). Although the frequency distribution of plasma 11-KT levels among males is often bimodal, the amount of overlap depends on when the fish were sampled (Larsen et al. 2004). Moreover, the assumption that classification of individual males based on springtime plasma 11-KT levels necessarily predicts their fall maturation status has not been directly verified. As part of an effort to better understand the genetic factors that influence minijack production, we are conducting a three-brood-year (BY; BY2014–2016) study to determine whether the age of the natural-origin broodstock used in a spring Chinook Salmon conservation hatchery influences minijack production among their progeny. This study requires unambiguous classification of each age-1 male spring Chinook Salmon either as immature or as a maturing minijack. Therefore, as an initial phase of this study, we tested a group of BY2013 juveniles and a subset of the BY2014 juveniles (1) to determine when to sample juveniles to obtain a bimodal plasma 11-KT frequency distribution, in order to establish a reliable cutoff value to distinguish between immature and maturing males; and (2) to verify that categorization of individuals based on the spring 11-KT cutoff value was in agreement with their maturation status during the following fall spawning season.

METHODS

Fish and study site.— The fish used in this study were the progeny of spring Chinook Salmon spawned in 2013 and 2014 at the Cle Elum Supplementation Research Facility (CESRF; Cle Elum, Washington), which is managed by the Yakama Nation. Broodstock at the CESRF are natural-origin anadromous fish captured at the Roza Adult Monitoring Facility located adjacent to the Roza Diversion Dam (river kilometer 206) on the Yakima River; captures occur primarily in May and June of each year during the adults' return migration to the upper Yakima River. The broodstock are transported to the CESRF and held until spawning in September and October.

Sampling of BY2013 fish involved a random sample of juveniles (N = 270) collected from the standard CESRF smolt production. These fish were lethally sampled on February 18 and February 19, 2015, in conjunction with pre-release disease screening. Rearing and sampling protocols and sample analyses were performed as described below for the BY2014 fish. The BY2014 juveniles were the product of a series of factorial test matings among a subset of the CESRF broodfish that included 29 females and 33 males, resulting in 87 unique families. Fertilized eggs from each family were incubated separately; in March 2015, the surviving swim-up fry were counted, and approximately 50 fry/family (total N = 4,350) were transferred to a raceway for rearing to the smolt stage in accordance with standard CESRF practices (see Fast et al. 2015). The raceway (30.5 m long × 3.5 m wide × 3.1 m deep) was partitioned to a length of 7.6 m and then 15.2 m as the fish grew, with 18 and 36 m^3 of water, respectively, being available to the fry. All fish were fed size-appropriate Bio-Oregon feed (www.bio-oregon.com) four times per day, with the feeding rate adjusted to obtain a target average smolt size of 152 mm and 40.0 g by March 15, 2016 (Legasse 1988). On April 4–7, 2016, most of the fish (N = 2,452 total; N = 1,254 males) were sacrificed in an overdose of buffered tricaine methanesulfonate (MS-222; 0.1 g/L; Western Chemical; www.wchemical.com) and then sampled to obtain data for the parent age effect study. For the present initial phase (aimed at monitoring development in individuals), a subset of 459 randomly selected juveniles were nonlethally anesthetized, measured, and sampled; injected with a passive integrated transponder (PIT) tag; and then returned to freshwater to recover. After recovery, the fish were transferred to one of four 500-L, circular fiberglass tanks for continued rearing. Fish were supplied with flow-through river water at ambient temperature, reared outdoors under natural photoperiod, and fed four times per day ad libitum. The tanks were checked daily for

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mortalities; when observed, the mortality date and PIT tag number were recorded. Surviving fish were lethally sampled on September 14, 2016, 1 week into the 5-week-long spawning season at the CESRF. We attempted to have a minimum sample size of 100 males at each sampling point to allow for clear delineation between maturity groups. All fish rearing and sampling procedures followed guidelines that were approved by the Institutional Animal Care and Use Committee at the University of Idaho.

Sampling procedures.— Lethal sampling consisted of (1) recording total body weight (BW) and FL; (2) taking a blood sample after caudal amputation using a heparinized Natelson tube; (3) visually identifying the phenotypic sex after dissection and inspection of the gonad; and (4) collecting a fin clip. During the February 2015 sampling of the BY2013 juveniles, 270 fish (125 males) were lethally sampled (FL [mean ± standard error of the mean {SEM}] = 115.7 ± 0.86 mm; BW = 17.9 ± 0.41 g). In April 2016, 2,452 of the BY2014 juveniles (1,254 males) were lethally sampled (FL = 153.0 ± 0.40 mm; BW = 41.0 ± 0.35 g). An additional 459 fish were nonlethally sampled (FL = 150.0 ± 0.50 mm; BW = 37.9 ± 0.46 g) using similar procedures except that blood samples were collected via caudal vein puncture using a heparinized 23-gauge, 12.7-mm (0.5-in) needle attached to a 1-mL syringe, and each fish was injected with a PIT tag into the peritoneal cavity. In September 2016, surviving fish from the nonlethally sampled group (N = 186; FL = 178.6 ± 1.03 mm; BW = 67.8 ± 1.42 g) were sacrificed, and their PIT tag numbers were recorded. They were subject to lethal sampling as outlined above, and the testes in males were weighed. Blood samples were immediately centrifuged at 9,400 × g for 5 min, and the plasma was aspirated and frozen on dry ice. Plasma samples were transported to the University of Idaho and stored at −80°C until laboratory analysis for 11-KT was performed.

Plasma sample processing.— Plasma 11-KT concentrations were assayed using a commercially available 11-KT enzyme-linked immunosorbent assay (ELISA) kit (Biosense; Cayman Chemical, Ann Arbor, Michigan). Plasma samples were diluted and assayed in triplicate in the ELISA according to the manufacturer's instructions. The intra-assay coefficient of variation (CV) was 3.2%, and the interassay CV was 8.2%. Prior to being used in the ELISA protocol, plasma samples were solvent extracted twice with diethyl ether (Caldwell et al. 2014). These assays and the extraction procedure have been validated for *Oncorhynchus* species (Nagler et al. 2012; Caldwell et al. 2014).

Genetic sex identification.— All fin clip samples were processed by the Columbia River Inter-Tribal Fish Commission's molecular genetics laboratory at the Hagerman Fish Culture Experiment Station (Hagerman, Idaho). Extraction of DNA from the fin tissue was accomplished using the Chelex 100 method (Sigma-Aldrich, St. Louis, Missouri). The DNA samples were then genotyped using genotyping-in-thousands by sequencing (GT-seq; Campbell et al. 2015). Genetic sex was determined using primers for the OtsY3 sequence discovered by Brunelli et al. (2008) and a protocol previously described by Hess et al. (2016). This is a Y-chromosome-specific assay that has been validated to differentiate genetic sex in Chinook Salmon (Brunelli et al. 2008; Campbell et al. 2015). For consistency with the larger parent age study and to remove any misidentifications made during the visual inspection for phenotypic sex, only data for the genetic males were used in the analyses presented. Data for fish that failed to meet genotyping thresholds (N = 8) were removed.

Data analyses.— Plasma 11-KT levels were $\log_{10}$ transformed prior to analyses, while FL and BW were $\log_e$ transformed. The gonadosomatic index (GSI) was calculated as

$$\left[\frac{\text{mass of testes (g)}}{\text{total body mass (g)}}\times 100\right].$$

Cutoff analyses were performed in R version 3.4.3 (www.R-project.org) utilizing Hartigans' dip test (part of the "diptest" package; CRAN.R-project.org/package=diptest), NormalMixEM, and cutoff probability codes (the latter two being part of the "mixtools" package; www.jstatsoft.org/v32/i06/) to determine (1) whether the distribution was indeed bimodal; (2), if so, what two-model normal curves best fit the distributions; and (3) the cutoff value that had a 50:50 chance of being selected from either curve. Cluster analyses of individual plasma 11-KT versus GSI data were conducted using Ward linkage, the data were standardized, and two clusters were specified. The effects of maturation status (immature male versus maturing male), FL, and the maturation status × FL interaction on BW were assessed using a generalized linear model (GLM) with a normal distribution and an identity link function. Comparisons of BW and FL between immature males and maturing males were made using t-tests performed in Prism version 7c (GraphPad Software, Inc.; www.graphpad.com). Unless otherwise noted, all analyses were performed in JMP version 13 (SAS Institute, Inc.; www.jmp.com). For all analyses, significance was evaluated at $\alpha = 0.05$, and values are expressed as mean ± SEM.

RESULTS

Examination of the plasma 11-KT values for BY2013 male fish sampled in February 2015 (N = 125) revealed a potentially bimodal distribution; however, Hartigans' dip test was not significant (test statistic $Dn = 0.032$, $P = 0.47$; Figure 1A). Nonetheless, analysis to determine a cutoff value was performed, yielding a value of 1.99 ng/mL. Analysis of the distribution of plasma 11-KT values for lethally sampled BY2014 male fish sampled in April 2016 (N = 1,254) revealed a significantly bimodal distribution

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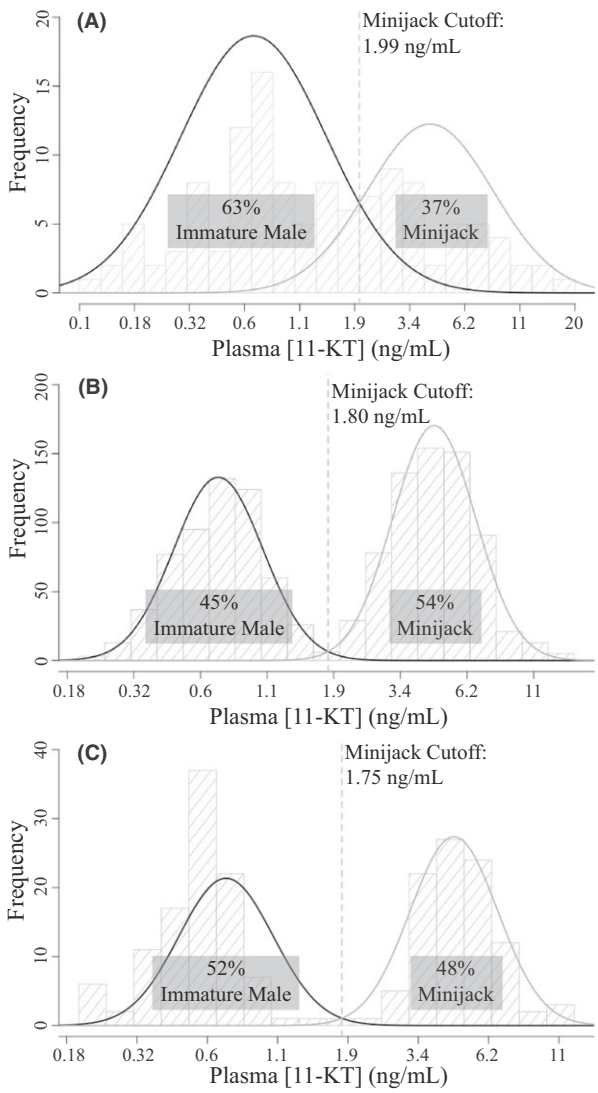

FIGURE 1. Frequency distribution of $\log_{10}$ transformed plasma 11-ketotestosterone (11-KT) levels from male age-1 spring Chinook Salmon sampled at the Cle Elum Supplementation Research Facility (Yakima River, Washington): **(A)** brood year (BY) 2013 fish (N = 125) that were lethally sampled in February 2015; **(B)** BY2014 fish (N = 1,254) that were lethally sampled in April 2016; and **(C)** BY2014 fish (N = 202) that were nonlethally sampled in April 2016. Solid lines indicate the best-fit normal distribution curves to the data assuming two populations. Dashed lines represent the 11-KT value with an equal probability of coming from either distribution. Hartigans' dip test performed on all three groups indicated that the distribution from the February sampling (panel A) was not significantly bimodal (P = 0.47), whereas the distributions from the April samplings (panels B and C) were both significantly bimodal (P < 0.001).

(Dn = 0.063, P < 0.0001). Analysis determined the cutoff value to be 1.80 ng/mL (Figure 1B). The same set of analyses performed on samples collected from the nonlethally sampled BY2014 males (N = 202) also revealed a similarly significant bimodal distribution (Dn = 0.12, P < 0.0001) and cutoff value (1.75 ng/mL; Figure 1C).

Cluster analysis of April and September plasma 11-KT and September GSI data for the nonlethally sampled males that survived to September (N = 103) separated the population into two distinct groups that were in 99% (102/103) agreement with the groups formed based on the plasma 11-KT analysis alone (Figure 2A). The exception was one fish with an April plasma 11-KT value (1.59 ng/mL) that fell slightly below the 1.75-ng/mL cutoff value; this individual was classified as a maturing male based on its September GSI as well as its high September plasma 11-KT value. Cluster analysis of September plasma 11-KT values and September GSIs from the surviving nonlethally sampled males also separated the population into two distinct groups for which there was 100% (103/103) agreement with the cutoff value analysis (Figure 2B).

Average plasma 11-KT values varied significantly between maturity groups and sampling time points (two-way ANOVA: F = 1,926, df = 3, P < 0.0001; Figure 3). Characterization of maturity groups was based on plasma 11-KT from male fish that survived until the September sampling point (N = 103), with data for the one fish whose maturity status was incorrectly predicted by its April plasma 11-KT removed from both sampling time points. In April, immature males (N = 38) had an average plasma 11-KT of 0.60 ± 0.035 ng/mL, whereas maturing males (N = 65) had an average of 5.0 ± 0.21 ng/mL. Between April and September, immature males experienced a significant fivefold increase in average plasma 11-KT (2.84 ± 0.69 ng/mL), while minijacks experienced nearly a 50-fold increase that resulted in an average plasma 11-KT level of 245.8 ± 14.47 ng/mL.

Analysis of BW (Figure 4A) and FL (Figure 4B) data for all males sampled in April 2016 revealed that the maturing males were significantly heavier (unpaired t-test: t = 25.24, df = 1,448, P < 0.0001) and longer (t-test: t = 22.23, df = 1,448, P < 0.0001) than their immature counterparts. A GLM investigating the effects of FL, maturity status (immature male or minijack), and the interaction of the two on BW (Figure 4C) revealed no effect of the maturity status × FL interaction (ANCOVA: χ^2 = 0.22, df = 1, P = 0.64); therefore, the interaction term was dropped, and the analysis was re-run. This analysis revealed that both FL (ANCOVA: χ^2 = 4,154.2, df = 1, P < 0.0001) and maturity status (ANCOVA: χ^2 = 207.6, df = 1, P < 0.0001) had significant effects on BW, indicating that for a given FL, maturing males were 5.4% heavier at the April sampling.

DISCUSSION

By following individual fish, this study provides strong evidence that plasma 11-KT measured in age-1 spring Chinook Salmon males at the time of smolt release accurately predicts their maturation status the following fall at

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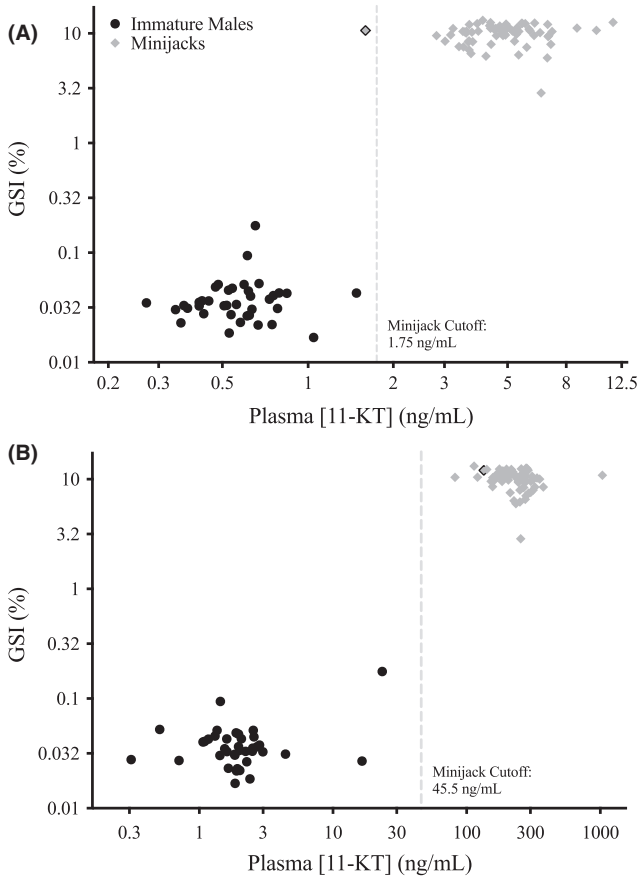

FIGURE 2. Cluster analysis of plasma 11-ketotestosterone (11-KT) levels versus gonadosomatic index (GSI) values in surviving male spring Chinook Salmon sampled at the Cle Elum Supplementation Research Facility (Yakima River, Washington): **(A)** April 2016 plasma 11-KT versus September 2016 GSI; and **(B)** September 2016 plasma 11-KT values versus September 2016 GSI (N = 103). The dashed line represents the 11-KT value with an equal probability of representing either maturity status from the bimodality analysis (1.75 ng/mL in April; 45.5 ng/mL in September). The gray diamond outlined in black indicates the one individual that was incorrectly identified as an immature male based on its April plasma 11-KT level.

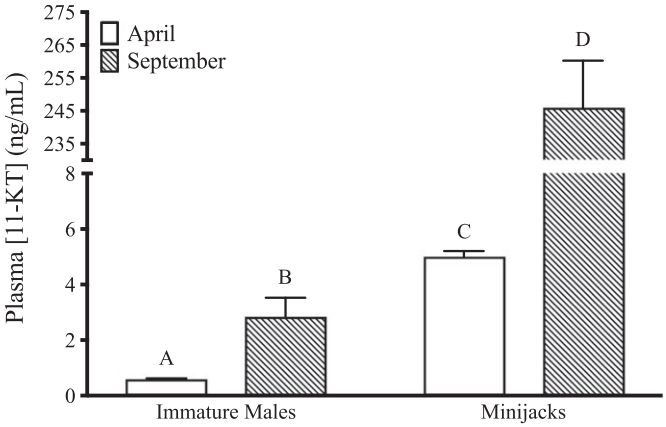

FIGURE 3. Mean (±SEM) plasma 11-ketotestosterone (11-KT) levels in immature and maturing male spring Chinook Salmon sampled at the Cle Elum Supplementation Research Facility (Yakima River, Washington) that survived between sampling points in April (open bars) and September (hatched bars; N = 103). Average plasma 11-KT values differed significantly ($P < 0.05$) between maturity groups and sampling time points, as indicated by the different letters above each bar.

age 2. The study is also unique in its incorporation of a statistical analysis for determining the bimodality of the plasma 11-KT frequency distribution and cutoff value between immature and maturing males. Doing so revealed that waiting until April to sample provided much better separation than sampling in February. Plasma 11-KT levels in February 2015 were not significantly bimodally distributed. Sampling in April 2016 produced a significant bimodal distribution of the plasma 11-KT values, with very little overlap between distributions, and a cutoff value that correctly identified the maturity status of male age-1 spring Chinook Salmon (based on September GSI) 99% of the time. Thus, although the results provide convincing evidence that springtime plasma 11-KT values accurately predict the maturation status of individual males during the following spawning season, delaying the sampling until the time of smolt release provides the clearest separation between the two groups.

The cutoff value obtained in our April sampling was comparable to values from previous studies conducted at the CESRF. In one of the first studies investigating plasma 11-KT as an indicator of minijack development in salmonids, Larsen et al. (2004) sampled BY2000 juvenile spring Chinook Salmon from the CESRF in March 2002 and deduced that the cutoff value was 0.8 ng/mL. Studies of spring and summer Chinook Salmon have demonstrated that plasma 11-KT progressively increases during maturation (Campbell et al. 2003; Larsen et al. 2004), consistent with the higher cutoff value found at a later sampling time in the present study. In addition, variation in laboratory evaluation methods (i.e., different extraction methods and assay protocols) would be expected to result in somewhat different values between laboratories. Although the cutoff values differ somewhat, the plasma 11-KT distribution histograms indicate that the rates of precocious male maturation were similar between studies. If one's goal is to classify male age-1 spring Chinook Salmon with near 100% accuracy as immature or maturing males, as was the case in our study, these considerations make it clear that cutoff values should be evaluated statistically in each cohort of fish and at each sampling time point. If instead the goal is simply to obtain an overall estimate of the rate of precocious male maturation in a population, one could use earlier sampling points, albeit with the resulting 11-KT distribution histograms showing more overlap between distributions and greater uncertainty in the estimated cutoff value.

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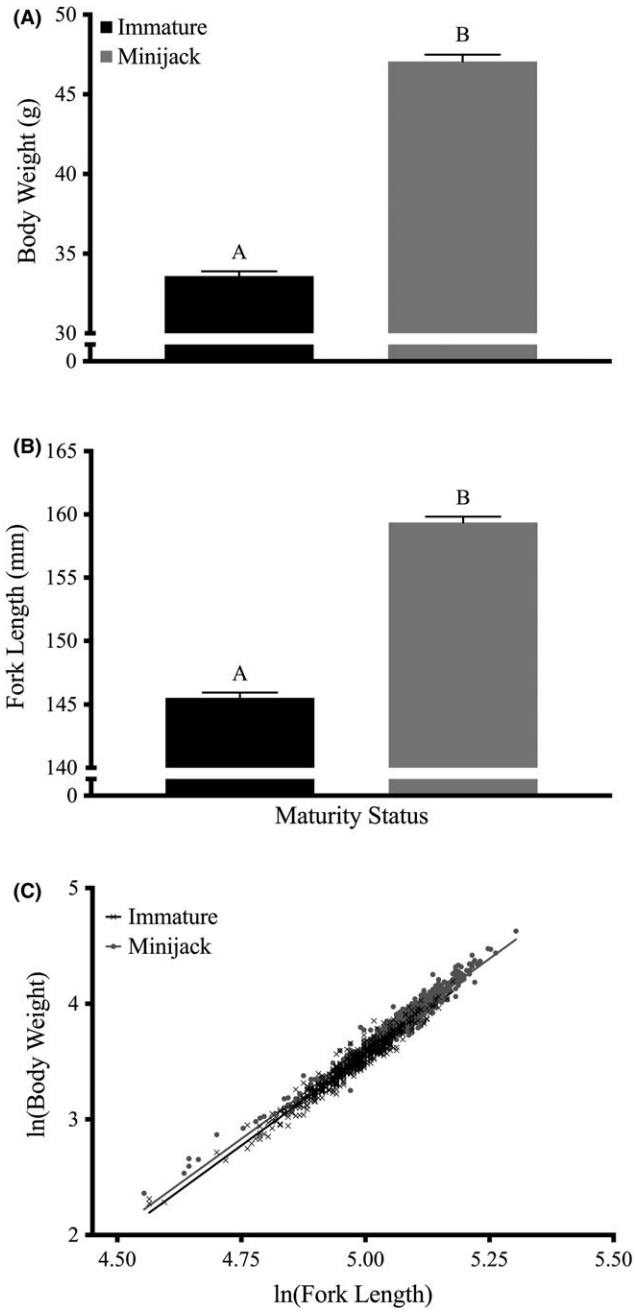

FIGURE 4. **(A)** April mean (±SEM) body weight, **(B)** April mean (±SEM) FL, and **(C)** April ln(FL) versus ln(weight) relationship for immature and precociously maturing male spring Chinook Salmon sampled at the Cle Elum Supplementation Research Facility (Yakima River, Washington). Maturing males were significantly heavier (panel A) and longer (panel B) than immature males. A generalized linear model analysis investigating the effects of FL, maturity status, and the interaction of the two on total body weight (panel C) revealed that there was no effect of the maturity status × FL interaction ($P = 0.99$). There was, however, a significant effect of both FL and maturity status on total body weight ($P < 0.0001$ for both variables). For any given FL, maturing males were 5.4% heavier than immature males during the April sampling. Different letters above bars represent a significant difference ($P < 0.05$) between groups.

Increases in GSI reliably indicate reproductive maturation as male salmonids approach spawning, and GSI is often used as a proxy for reproductive effort (Unwin et al. 1999; Quinn 2005; Butts et al. 2012). The September GSI values observed in maturing males in this study are comparable with those observed in other studies on age-2 precociously maturing male Chinook Salmon, indicating that these fish were indeed completing maturation (Unwin et al. 1999, 2005). Maturity status based on 11-KT and GSI was further supported by visual examination of the testes; fish with low GSI and 11-KT values had testes that were thin and translucent pink in color, while the testes of fish with high values were large and opaque white, with actively flowing milt in many of them. These observations were similar to the morphometric quantifications used by Larsen et al. (2004) to categorize age-1 upper Yakima River basin spring Chinook Salmon into groups based on maturity status. Moreover, there was a nearly complete concordance between fish identified as maturing males based on elevated April 11-KT levels and fish that completed maturation based on September GSIs. This indicates that April plasma 11-KT can be relied upon as an accurate predictor of individual maturation status the following September.

We did not observe any individuals with resorbed gonads. There was relatively little variation in GSI, and all GSI values were in or near ranges previously reported for precocious male and jack male Chinook Salmon (Unwin et al. 1999; Butts et al. 2012), providing no support for occurrence of arrested maturation in any of the study fish. This is consistent with the hypothesis that our fish had passed the spring critical period and were committed to either completing maturation or remaining immature (Thorpe et al. 1998). However, our study was conducted under hatchery conditions with continued feeding. Juveniles reared at the CESRF are typically released into the upper Yakima River, and food resources are restricted during smolt out-migration. We cannot exclude the possibility that under these conditions, juveniles exhibiting high 11-KT prior to release might arrest gonadal development. However, we do not believe that this is likely, as anadromous salmon complete maturation between the time of re-entry into freshwater and the subsequent spawning season—a period during which the fish naturally do not feed (Quinn 2005). Out of the 103 male fish that survived until the September sampling, only one was incorrectly classified as immature based on its April plasma 11-KT value, which (at 1.58 ng/mL) was the closest individual value in the immature distribution to the 1.75-ng/mL cutoff. In September, among the 38 immature male fish, there were two that had plasma 11-KT levels significantly elevated relative to the others. A similar occurrence has been observed in other studies (D. Larsen, National Oceanic and Atmospheric Administration

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Fisheries, personal communication) and possibly identifies individuals that are initiating maturation and, were they permitted out-migration to the ocean, would return the following year as 3-year-old jacks.

Substantial mortality (58.6%) occurred in the nonlethally sampled juveniles held for continued rearing from April to September. The mortality was no doubt associated with stress and injury from the handling required to collect size measurements, a fin clip, and a blood sample from each, plus insertion of a PIT tag. However, we do not believe that this compromises our conclusion that categorization of males by April 11-KT level is accurate. For the mortality to introduce bias, any high 11-KT males that were actually going to arrest maturation and any low 11-KT males that actually were going to belatedly complete maturation would have had to die. If such individuals were present, they would be expected to have 11-KT values close to the region where the two distributions overlap. This was not the case; mortality was not associated with 11-KT level in either group (data not shown). The mortality strengthens our conclusion that males did not arrest maturation, insofar as the associated stress and reduced feeding would be expected to affect survivors. Nonetheless, the mortality indicates that the combination of caudal vein puncture, PIT-tagging, and other procedures during the April sampling is not optimal as a method to nonlethally assess maturation status in fish that are slated to be released.

Both maturing and non-maturing males experienced a significant increase in 11-KT level from April to September. The increase in maturing males can be attributed to the process of testis maturation. The precocious males in this study experienced a 50-fold increase in plasma 11-KT over the 5-month period, with the September levels averaging 245.8 ± 14.5 ng/mL. These levels are comparable to —though somewhat higher than—previously reported levels for both precocious and normal aged mature spawning male Chinook Salmon, which generally were in the 150–200-ng/mL range (Campbell et al. 2003; Butts et al. 2012). In the current investigation, the mean GSI for the precociously maturing males was 10.0 ± 0.23%, while that for anadromous adult male Chinook Salmon averaged 5.5 ± 0.28% (Butts et al. 2012). This difference suggests that precocious males commit a greater portion of energetic investment to germ cell development relative to somatic tissue growth compared with anadromous males. This higher relative milt volume may provide an advantage, or at least an equalizing effect, for the sneaker mating strategy when precocious males are competing with much-larger anadromous males during spawning (Quinn 2005; Unwin et al. 2005; Butts et al. 2012).

Plasma 11-KT levels also increased significantly between April and September in immature males. Although the two immature fish that had notably higher 11-KT levels than the rest of the immature males contributed to this increase, the difference remained significant when data for these individuals were excluded. A similar September increase in plasma 11-KT level in immature males has been reported previously (Campbell et al. 2003). These observations suggest that even in immature male Chinook Salmon that will not complete the maturation process for at least another year and up to 4 years, the reproductive endocrine axis is activated in September, when mature individuals are spawning. Previous studies in salmonids and other teleosts have reported increases in reproductive hormone levels and even partial development of gametes in pre-pubertal fish (Hassin et al. 2000; Wylie et al. 2017), the timing of which coincides with sexually maturing individuals. It has been postulated that these "dummy runs" exhibited by pre-pubertal individuals are a reflection of a functioning brain–pituitary axis, which responds to the appropriate environmental conditions, but underdeveloped gonads that do not respond properly to stimulation by their respective secretagogues (Hassin et al. 2000; Okuzawa 2002; Wylie et al. 2017). The timing of the fall increase in plasma 11-KT levels in immature males is consistent with the proposed critical period for initiation of maturation in fall-spawning salmonids (Mangel 1994; Thorpe et al. 1998), suggesting that activation of the reproductive endocrine axis at this time may be associated with the decision to initiate maturation. To develop strategies to reduce the incidence of minijacks, future investigations are needed to better define the timing of the critical decision windows and to determine which physiological factors stimulate precocious maturation.

Hatchery rearing regimes for spring Chinook Salmon are designed to produce fast-growing, large smolts for release in the spring because larger smolts are more likely to survive out-migration to the ocean and to return at higher rates as anadromous adults (Martin and Wertheimer 1989; Zabel and Achord 2004; Piche et al. 2008). However, elevated growth in the hatchery and a larger size at release are also associated with increased numbers of minijacks (Larsen et al. 2006, 2013). Corroborating previous results, we observed maturing males sampled in April, coincident with the time of smolt release, to be 5.4% heavier at a given length than immature males. This implies that the growth history of immature and precociously maturing males diverged at some point prior to this; however, it does not provide sufficient resolution on when the differential growth occurred to draw conclusions about how growth and maturation interact in the minijack life history. The critical period hypothesis of salmonid maturation predicts that minijack growth, size, and/or adiposity would have been elevated during the proposed initiation of maturation period the previous fall (Mangel 1994; Thorpe 1994; Thorpe et al. 1998). However, in addition, salmonid appetite and growth are increased during the early stages of maturation, suggesting that growth

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may have been elevated from the initiation of maturation to sampling in April (Stead et al. 1999; Campbell et al. 2003; Taranger et al. 2010). Further study is required to evaluate whether differential growth occurs over either or both of these time periods.

In summary, this study provides compelling evidence that springtime plasma 11-KT levels can be used to accurately predict fall maturation status in individual fish. However, caution must be used when choosing a sampling date to ensure that the plasma 11-KT frequency distribution will produce a significantly bimodal distribution from which an accurate cutoff value can be determined. Plasma 11-KT cutoff values should be established for each cohort of fish and each sampling time.

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